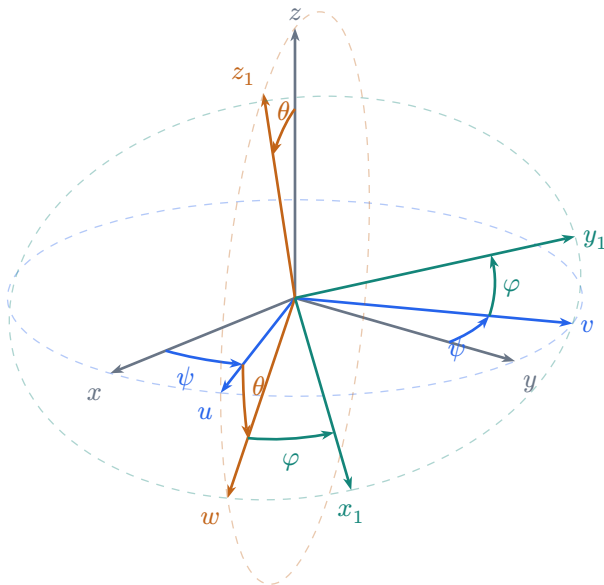


Euler angles: three ordered rotations

Intrinsic Z-Y-Z convention · axes move with the frame



1. Precession ψ

Rotate about z

2. Nutation θ

Tilt about v

3. Spin φ

Rotate about z_1

$$R = R_z(\psi)R_y(\theta)R_z(\varphi)$$

The final basis is (x_1, y_1, z_1) . Changing the rotation order changes the orientation.